



2E10 Final Report (Part A)

Group X15

Our team's performance, although just barely not good enough to attempt the gold challenge, was still solid. This is mainly due to the strengths leveraged by our team, three of which I will discuss in this section of the report.

The first and most significant one of these strengths was our team's use of effective communication. Our successes have been mostly attributed to this, as it's been clear since the beginning of the project that each member was proficient in and willing to communicate properly. This allowed us to keep our team on track at all times and give everyone access to the information regarding the progress made, which made our time in the labs more productive as individual members were not reliant on another person to catch them up on the current state of the project. This was all facilitated by a regularly used whatsapp group chat, as well as google drive, which was updated with all work done in real time.

Individual participation and responsibility were also major strengths of our team, and contributed massively to our team's successes. This means that each member undertook tasks and ensured that they were completed to the required standard and within the suggested time frame. Our proficiency in communication made this more efficient, as everyone was aware of the tasks being undertaken by others and could work on their own accordingly. This allowed for the project to be completed in a smooth manner, as it was rare for vital tasks to be left uncompleted and consequently bottle neck our progress.

Individual skills were also leveraged to their full potential, which allowed us to curb the difficulty of some of the more difficult tasks. Our team quickly established these unique abilities, such as one member being particularly proficient in coding, and others being better at problem solving, debugging, or circuit design. Although everybody in the group got a taste of every part of the project, it was still clear that these individual skills were leveraged as much as reasonably possible. The main benefit of this was the time and energy saved in not having to bring anybody's skills up to the required standard, as well as the more advanced knowledge possessed by individuals which made debugging and problem solving easier.

Despite our team having some weaknesses, it's fair to suggest that none of them are to blame for our inability to move onto and complete the gold challenge. However, we will still outline some aspects of the project's completion which could have potentially been organised or completed to a better standard.

The first of these aspects regards the distribution of work during periods of the project. With the required circuitry being completed before the bronze challenges demonstration, this aspect of the project saw no necessary progress for silver. As such, we were left with two main aspects of the project remaining: the arduino code and the processing code. Even though the processing and GUI sections of the project required significant effort during both the bronze and silver challenges, it was less than that required by the arduino code. This meant that even after everyone had completed all of their assigned tasks, we were left to debug the arduino code for a very long time. We spent the entirety of week 9, 10, and 11 debugging and refining the code according to the buggies behaviour, which meant that the time spent on all other aspects became far less significant relative to that spent tuning the buggy.

Another aspect which caused some slight issues was our poor delegation of tasks as outlined in our project plans. Each plan suggested that members were to usually undertake a single task, which would sometimes take either very little or a lot of time and effort to complete. For example, in our plan for the bronze challenge we assigned someone the task of "wiring the IR sensors", and in the plan for the silver challenge we assigned someone the task of "required wiring/circuitry", both of which took little to no time at all. On the other hand, we assigned people to either complete the entirety of the processing on their own, or for silver we assigned a single person to do all of the PID implementation, both of which were disproportionately large workloads. As such, we as a team had to actively assign and take responsibility for tasks as the project was completed, which although added to our groups flexibility, hindered our ability to evenly distribute workloads at some points in the project.

Another area which proved to be a weakness was the amount of lab time missed by members. It wasn't uncommon for a member of the group to be missing from the laboratory session, and although this did not prove to be a significant hindrance - especially since they would inform the rest of the group prior to their absence - it still decreased the productivity of these lab sessions which made debugging issues surrounding the buggies on-track behaviour slower and more difficult.

Since our team achieved silver outside of the time which would have allowed us to move on to gold, we will discuss what changes we would make to allow us to achieve gold should we have been given two more weeks to work on the project.

Firstly, we would have been stricter in our delegation of tasks for the silver challenge. This includes assigning individual members specific tasks to have completed for the Monday after the reading week, as we decided at the time that it would be best to leave those tasks up in the air, and just have people complete them as and if they saw fit. This ultimately did not work out particularly well and resulted in the labs of week 8 being quite unproductive, as everyone was working on individual tasks and the buggies behaviour couldn't be tested. As such, we barely missed out on passing silver in time to move on to gold, as a functions implementation in our object following mode was overlooked and not properly implemented. Should we have had more productive labs where we spent our time testing and debugging the buggy instead of writing code, we very likely would have passed silver in time to attempt gold.

Another change we likely would have made is concerned with the tuning of our PID. For silver, we tuned the PID by means of trial and error, which was extremely time consuming. Should we have had two more weeks, we would have probably attempted to graph outputs for different values and optimize the PID that way. This would have saved us time, as well as removed the PID as a significant variable in our tests, as we would have been more sure that our values were correct.

Finally, we would attempt to remove hardware issues as a variable in our testing. Throughout bronze and silver, we had been using two different sets of batteries - the rechargeable batteries provided (7.2V) and our own un-rechargeable AA batteries (9V). These two sets of batteries displayed different behaviours at the same PWM values, and as such switching between them required us to change these such values. In the end, we used the 7.2V batteries provided, and tried to keep them at a high level of charge to preserve consistent buggy performance. This switching between batteries ended up proving itself to be a huge waste of our time. As well as the batteries, we would have tuned the IR sensors earlier into the silver challenge. Prior to our first demonstration, we had not altered the sensitivity of the IR sensors at all, but when we did it made a significant difference in the buggies responsiveness. As such, we would have made this change earlier, which would have indefinitely increased our probability of achieving gold.

Ethical issues which we encountered were few to none. Since we only achieved silver we didn't encounter any issues with data collected by the camera, and otherwise there were no real ethical issues surrounding our specific use of the buggy. As such, it makes sense to extrapolate the functionalities of the buggy into a broader context and suggest how they could in fact cause ethical concern.

Firstly, we will go over the obvious privacy concerns that arise regarding the use of the camera. Although the image data isn't stored, it could still be of concern to the people in the buggies vicinity, especially since it's a private space and people have to consent to having their image taken. Should systems like this be used in other technologies, these concerns would be way more serious, especially if video or image data was being stored for more than a fraction of a second.

If you further expand on the core features of the buggy, you can see how they could be used for unethical means. The combination of distance sensing and path-following could theoretically be used in applications where autonomous movement is used for unethical activities (e.g., spying, delivery of illegal goods). It is also reasonable to suggest that similar technologies could be used in surveillance systems or military robotics, which also raise serious ethical concerns.

It is also important to mention the concerns surrounding the environment, in the form of the electronic and plastic waste associated with the buggy. For example, our buggy required a new arduino board, a new DC adapter, as well as the use of many batteries. Since many groups required a greater number of part replacements than ours, it's valid to suggest that this waste raises legitimate ethical concerns for the environment.